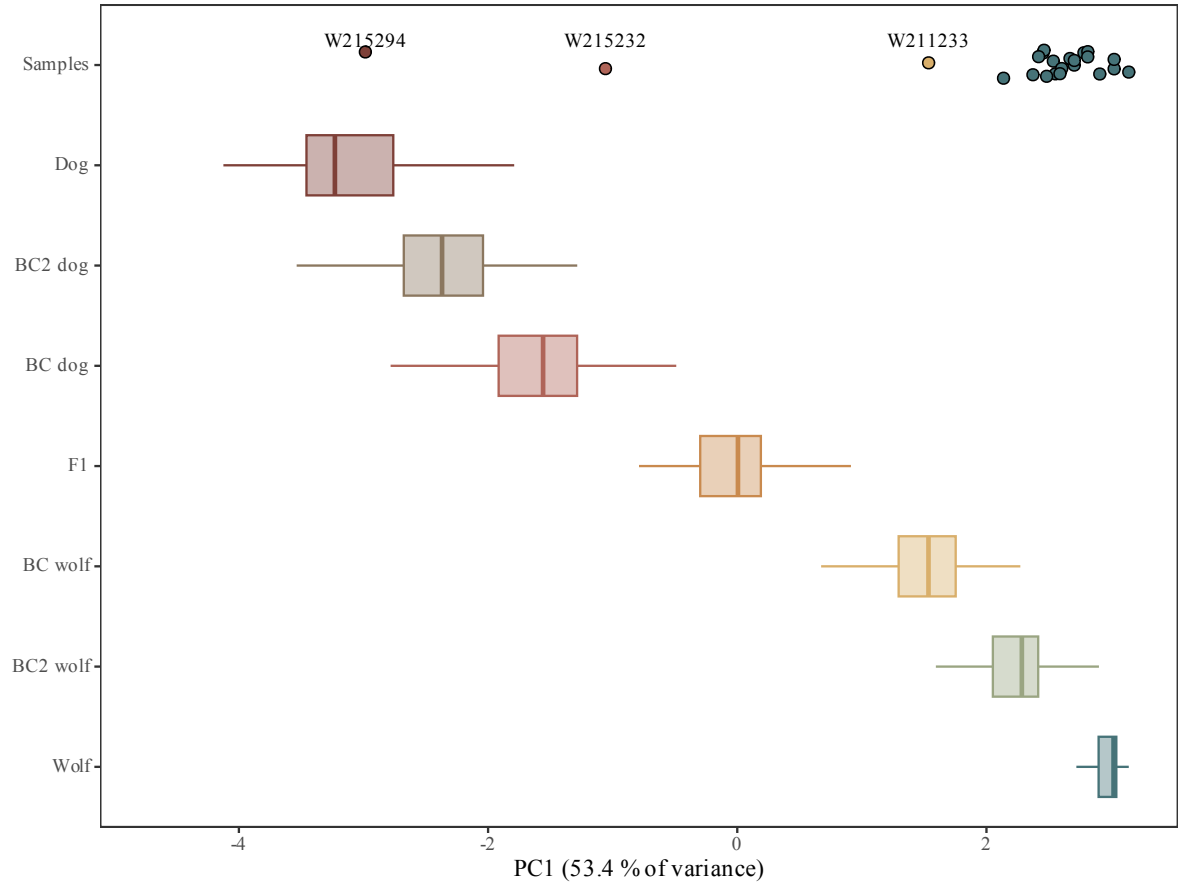PC1 of every simulated group and the samples